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Propagation and Insurance in Village Networks†

By Cynthia Kinnan, Krislert Samphantharak, Robert Townsend, and Diego Vera-Cossio*

Firms in developing countries are embedded in supply chains and labor networks. These linkages may propagate or attenuate shocks. Using panel data from Thai villages, we document three facts: as households facing idiosyncratic shocks adjust their production, these shocks propagate to other households on both the production and consumption sides; propagation is greater via labor than supply chain links; and shocks in denser networks and to more central households propagate more, while access to formal or informal insurance reduces propagation. Social benefits from expanding safety nets may be higher than private benefits. (*JEL* D13, D22, G22, I10, L14, L26, O10)

Most small enterprises in low- and middle-income countries (LMICs) are owned by individual, un-hedged households. The resulting lack of separation between production and consumption means that shocks to household spending needs can lead to production-side adjustments (LaFave and Thomas 2016). These adjustments may in turn affect other firms, propagating shocks through local production networks. Additionally, because households in LMICs often engage simultaneously in self-employment (running a firm) and wage work (working for others' firms), these production networks go beyond supply chains. They also involve firm-to-firm linkages through a labor market network of employer-employee transactions. As shocks propagate, there may be both production- and consumption-side indirect effects on the owners of other, linked, firms. Considering these responses (direct and indirect, and production- and consumption-side) together, as well as the mechanisms behind

* Kinnan: Tufts University and NBER (email: cynthia.kinnan@tufts.edu); Samphantharak: University of California San Diego, Puey Ungphakorn Institute for Economic Research (PIER), and Bank of Thailand (email: krislert@ucsd.edu); Townsend: Massachusetts Institute of Technology and NBER (email: rtownsen@mit.edu); Vera-Cossio: Inter-American Development Bank (email: diegove@iadb.org). Emi Nakamura was the coeditor for this article. We thank numerous colleagues and seminar audiences for helpful suggestions. Townsend gratefully acknowledges research support from the Thai Chamber of Commerce, the Thailand Research Fund, the Bank of Thailand, the Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD) (grant R01 HD027638), the Bill & Melinda Gates Foundation (grant 51935), and Private Enterprise Development in Low-Income Countries (PEDL) (funded by the Centre for Economic Policy Research and the Department for International Development under grant MRG002_1255). Opinions, findings, conclusions, and recommendations expressed here are those of the authors and do not necessarily reflect the views of the Bank of Thailand or the Inter-American Development Bank.

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them (propagation through supply chains or labor networks), is crucial for a full understanding of the welfare consequences of shocks and of safety nets.

We shed light on these issues, documenting three key facts. First, we empirically show that health shocks to households lead to direct production-side adjustments, which in turn propagate through local networks, affecting linked household firms in terms of production-side outcomes (e.g., local transactions, income) as well as consumption. Second, we distinguish propagation via supply chains versus via less-studied labor networks, and show that the extent of labor network propagation is greater. Third, we examine how the economic environment (network structure, access to formal and informal insurance, and frictions in labor markets) determines to what extent shocks are amplified or attenuated.

The setting we study is uniquely suited to answer these questions. The Townsend Thai data, constructed from 14 years of monthly panel survey data on households in 16 rural and peri-urban Thai villages, contain detailed information regarding intra-village transactions and transfers among family-operated businesses. We use these data to construct employer-employee (labor) and supply chain (goods) networks. We also have detailed information on firm performance and household consumption. The data additionally allow us to identify large, idiosyncratic shocks in the form of changes in health status, and to construct a valid counterfactual to obtain causal estimates of these shocks' effects. These elements together provide an ideal setting to shed light on the role of networks in propagating and mitigating shocks.

We show that an idiosyncratic shock has a significant effect on the business activities of the directly affected household. We follow Fadlon and Nielsen (2019) and compare the changes in outcomes of shocked households before versus after the shock, to those of other "control" households, who will experience a similar shock, but later on.[1] Shocked households are able to smooth consumption (nonhealth household spending). However, this smoothing is achieved by reducing business spending (by 23 percent), and almost entirely cutting demand for external labor (by 82 percent).[2] This finding validates that in our setting, separation between the consumption and production sides of households' balance sheets fails to hold.

A lack of consumption-production separation has also been found in other settings (see LaFave and Thomas 2016; Jones et al. 2022). Our contribution is to document three main findings. First, we leverage variation in the proximity of a given household to the shocked household through preshock networks to causally identify indirect impacts via propagation. We compare changes in outcomes before versus after the shock, between those more exposed (closer to the shocked household in the preshock network) and less exposed (farther away). Thus, we are able to quantify propagation at the micro level and trace its evolution within a network (village).

We find that idiosyncratic shocks propagate on both the production and consumption sides. Households with greater indirect exposure see larger falls in total transactions (a 19 percent decline for a unit change in closeness), and falls in income

[1] The outcomes of these "later shocked" households inform the counterfactual outcomes of the shocked households. For more discussion of how we construct the control group and the placebo shocks assigned to the control group, see Section II.

[2] The effect sizes are computed by diving the point estimate in Table 1 by the preshock mean of the dependent variable.

and consumption (7 percent and 3 percent declines for a unit change in closeness, respectively). The indirect effects are largest for households located one link away from the shocked households. However, these first-order indirect effects also propagate further, affecting households located two or more links away from the shocked household.

The presence of negative indirect effects demonstrates that the strategies used to smooth consumption against direct shocks (cutting business spending on goods and labor) are not only costly from a private perspective—the business scale of the shocked household is slow to recover—but also from a social perspective: the transactions, income and consumption of other households are also depressed. To the best of our knowledge, documenting *consumption-side* effects of firm-to-firm propagation is new in the literature.

We also provide evidence on how shocks propagate differentially through supply chain and employer-employee networks. We show that the propagation effect is stronger in labor market networks. While inputs and outputs can also be traded outside the village, the local nature of labor markets and the high monetary and nonpecuniary costs of temporary migration (Lagakos, Mobarak, and Waugh 2022) may prevent indirectly shocked households from adapting to the labor market disruptions caused by health shocks. This finding is consistent with the theoretical insights of Baqaee and Farhi (2020) that propagation is more severe when inputs are not substitutable, and demonstrates that the effects of propagation via supply chain networks cannot be directly extrapolated to propagation via other linkages. While the propagation role of supply chains in advanced economies is increasingly well understood (Carvalho et al. 2021), the role of labor market networks has not, to our knowledge, been studied.

To contextualize our results, we show that propagation occurs in the context of rigid local networks, in which the links between households exhibit a substantial level of persistence. Indirectly shocked households do not appear to replace "broken" links even two years after a shock. Existing links are vulnerable to shocks, and replacing these links appears challenging: suppliers struggle to find new customers when a client suffers a shock, and workers cannot easily find new jobs when existing employers scale back demand. As shocks propagate through village networks, their effects have important aggregate consequences. A back-of-the-envelope calculation yields a multiplier of 1.2.

We additionally show how features of the economic environment affect the indirect consequences of shocks. We first focus on whether shocks have larger indirect effects in denser networks. It is a priori ambiguous how the density of a network will affect propagation: on one hand, households interacting in denser networks may be able to replace broken links with the shocked household more quickly and easily. However, they may also be more exposed to the indirect effects of such shocks. Using cross-village variation in density across the 16 village networks in our data, we find that the indirect effects of idiosyncratic shocks on the *average* nonshocked household are amplified in denser networks, even after controlling for network size. Likewise, shocks to more central households—those with more links to other households before the shock—lead to larger negative indirect effects. These results suggest that network structure mediates the extent to which idiosyncratic negative shocks can translate into aggregate demand shocks, and that policies aiming to minimize aggregate fluctuations due to idiosyncratic shocks should be targeted either at more interconnected local economies or at central households.[3] These findings suggest a tradeoff: while increasing linkages among households may strengthen the insurance capacity of networks (Feigenberg, Field, and Pande 2013), increased links may also decrease resilience to propagation. Likewise, encouraging links with central households may promote information diffusion (Beaman et al. 2021), but may increase shock propagation as well.

We provide suggestive evidence on the roles of formal and informal insurance in attenuating propagation. Shocks to households who are better insured, formally or informally, propagate less. Access to insurance for directly shocked households can prevent costly production-side actions which drive propagation. These results underscore the distinct roles of village networks: supply chain and labor networks contribute to propagation, while insurance networks contribute to attenuation.

Our work contributes to the literature analyzing the aggregate implications of microeconomic shocks. Existing evidence largely focuses on the propagation of sectoral shocks via supply chain networks among larger firms in high-income countries.[4,5] In LMICs, small, family-operated firms (i) make intertwined consumption and production decisions; (ii) are exposed to shocks not typically faced by large firms (e.g, shocks to family health and wealth); and (iii) participate in multiple networks. A unique setting allows us to document the consequences of these shocks for the local economy. In our setting, households smooth consumption amid idiosyncratic shocks by making costly production-side adjustments, which, in turn, negatively affect the consumption levels of other households.

Finally, our results provide insights related to the mechanisms behind multiplier effects: a change in aggregate activity greater than the direct effect alone. Several studies document multiplier effects arising from large inflows of cash into local economies in both high- and low-income countries (Nakamura and Steinsson 2014; Chodorow-Reich 2019; Egger et al. 2021). Our approach enables us to contribute with a novel insight: *idiosyncratic* shocks can amplify and generate multiplier effects similar to those generated by large inflows of resources. In addition, while recent work in LMIC contexts identifies equilibrium channels such as wages (see, Egger et al. 2021; Franklin et al. 2021; Breza and Kinnan 2021) and prices (Burke, Bergquist, and Miguel 2019; Cunha, De Giorgi, and Jayachandran 2019) as important for indirect effects, less is known about locally heterogeneous exposure to indirect effects via supply chainsupply chain and labor networks.[6] Detailed data on

[3] Allen and Gale (2000) and Elliott, Golub, and Jackson (2014) examine the role of network structure theoretically in the context of links between financial intermediaries while Bigio and La'o (2020) examine the case of input-output networks. To the best of our knowledge the role of network structure in propagation among smaller firms has not previously been examined nor have these effects been tested empirically.

[4] See for instance, Carvalho et al. (2021); Barrot and Sauvagnat (2016); Caliendo et al. (2017); Dhyne et al. (2021); Huneeus (2019).

[5] Also related are several studies of the adjustment of larger, formal firms in LMICs to shocks, Khanna, Morales, and Pandalai-Nayar (2022), who examine how supply chain networks among registered (i.e., larger) establishments within an Indian state respond to the aggregate shock of COVID-19 border closures, and Felix (2022) who studies the labor market consequences of trade exposure in Brazil for formal sector workers.

[6] Angelucci and De Giorgi (2009) and Moscona and Seck (2021) show that cash transfers via public programs spill over to noneligible households via risk-sharing networks; their focus is not on production-side network spillovers, nor on shocks that are prima facie idiosyncratic to individual households.

networks enables us to estimate within-village spillovers that are *net* of any effects on wages or prices; a crucial distinction for policy (Guren et al. 2021).

I. Context and Data

A. *Household Data*

Our data come from the Townsend Thai Monthly Survey, which covers 16 villages in northern and central Thailand. The sample includes approximately 45 households per village, representing on average 42 percent of the village population (Townsend 2010). A baseline interview was conducted from July to August 1998, collecting information on the demographic and financial situation of the households. Monthly updates began in September 1998. We complement these data with data on financial accounts of the households in the sample (Kinnan et al. 2023a).[7] In both cases, we use data which covers the period between September 1998 and December 2012 and focus our analysis on the subset of 510 households who responded to the interview throughout all survey waves.

Panels A and B of online Appendix Table A1 characterize sample households in terms of demographic and financial characteristics. All households in the sample operate a family firm. Households derive income mostly from family farms, but they also operate off-farm businesses. At the same time, they supply labor to other firms in the village or other external employers. In addition, part of total income comes from the receipt of government transfers and/or gifts from other households.

B. *Network Data*

We complement the monthly panel with detailed information on transactions among households capturing different economic links (Kinnan et al. 2023b). In each survey wave, interviewees identify all households in the village with whom they have conducted a given type of transaction.[8] We aggregate the monthly transactions by year to elicit three types of networks—supply chain, labor, and financial—for each year in the sample. The fact that households operate firms, sell and buy labor to other firms, and provide and receive transfers to and from other households in their village, defines three types of firm-to-firm linkages: input and output (supply chain), employer-employee (labor market), and gifts (financial). Specifically, supply chain networks capture transactions of final and intermediate goods across firms operated by households in the same village. Labor networks capture cross-household labor flows within the village.[9] Financial networks are defined by gifts and loans across households in the same village. We also construct kinship networks, which

[7] For more detail about the survey, see Samphantharak and Townsend (2010).

[8] The set of transactions includes relinquishment of assets, purchases or sales of inputs or final goods, provision of paid and unpaid labor, and giving and receiving gifts and loans. As is typically the case in networks based on survey data, our networks may look thinner than those that would be elicited using census data (Chandrasekhar and Lewis 2016). We discuss the implications for our research design in Section III.

[9] Outgoing labor flows (labor sales) arise when some members work in the household business and others work for other local businesses, or when the same household member(s) divide their time between the household business and working in local labor markets. Incoming labor flows (labor purchases) arise when household businesses hire from the local labor market.

are measured in the initial survey wave. Online Appendix Figure A1 depicts these networks for one sample village in one year.

Panel C of online Appendix Table A1 summarizes network participation across the sample as a whole. Just over one-half (51 percent) of the households transact in the local village supply chain network by trading inputs and final outputs, with 1.36 connections on average per year; and 66 percent exchange labor with other households in the village, with 3.33 connections on average. An average of 38 percent of households participate in the financial (i.e., gift/loan) village networks, and 77 percent of households have at least one relative in the village.

Households participate in several networks within a given period. For those linked through gift/loan networks, over 60 percent also transact in supply chain networks and over 76 percent of them transact in labor markets. Over 77 percent of households transacting in the village supply chain networks also sell/purchase labor locally, and 46 percent are linked through local financial markets.

Online Appendix Table A1 panel D concerns the size of sample villages and firms. The average village has 161 households. There is evidence of excess kurtosis in the firm size distribution (measured via gross revenues): average village-level kurtosis is 10 (excess kurtosis of 7). Following Gabaix (2011), excess kurtosis suggests that idiosyncratic shocks may have important aggregate effects in our setting.

C. *Constructing Idiosyncratic Shocks*

To understand how shocks propagate to other households through village networks, we focus on idiosyncratic events associated with high levels of health spending, which correspond to periods of high financial stress. These shocks are well-suited for our analysis for several reasons. First, serious health shocks affect household finances and labor supply (Gertler and Gruber 2002; Genoni 2012; Hendren, Shenoy, and Townsend 2018). The large magnitude of such shocks improves statistical power and moreover such shocks are of prima facie importance. Second, because these shocks are uncorrelated across households (as shown below), we are able to separate the direct idiosyncratic effect from indirect effects hitting other connected households via propagation. Additionally, the timing of these shocks, as we show below, is plausibly exogenous.

We identify shocks as follows (see online Appendix B for details). For each household, we identify the month with the highest level of health spending over the panel.[10] We focus on the largest shocks, because they pose a significant financial burden to the household. To account for potential anticipation effects, we define the beginning of an event by subtracting the number of months preceding the episode of high health spending during which household members reported health symptoms from the month corresponding to the episode. Thus, although we use health spending data to identify shocks, the *timing* of the onset of the shock is coded based on changes in health status.[11]

We restrict this search to health spending episodes occurring during years 3–12 of the panel (out of 14 years). This enables us to observe at least 2 years both

[10] Thailand has a universal health insurance program covering (some) health costs, so these expenses are above and beyond those covered; see online Appendix C.

[11] Online Appendix Figure A8 shows that, prior to the sudden increase in health spending, the median number of consecutive months in which households report any health symptoms is three. More details are in online Appendix B.

pre- and postshock. By construction, we identify one shock per household, ensuring that the propensity of experiencing a shock is not correlated with characteristics such as health, income or network position. We exclude shocks related to childbirth, which may be planned/anticipated before the onset of any symptoms, leaving 476 shocks.

D. *Characteristics of the Shocks*

Relationship between Health Spending and Health Status.—A natural question is how our spending-based shock measure correlates with changes in health status. Online Appendix Figure B2 shows that self-reported symptoms co-move with health spending, confirming that shocks are identifying decreases in household health. Online Appendix Figure B3 reports the distribution of behaviors and symptoms reported by shocked households during the year following a shock and during nonshock periods. Usage of health facilities, particularly hospitalization, is substantially higher after shocks, and there is a greater likelihood of a household member suspending their daily activities for 1, 7, 14, or 21 days, consistent with shocks capturing sharp changes in health. In addition there is a higher incidence of uncommon symptoms (which tend to be more severe in this context).

Magnitude of the Shock.—These shocks represent a substantial financial burden to affected households. On average, the highest level of monthly health spending within six months after the onset of the shock (7,647 baht) accounted for 99 percent of the average monthly consumption during the six months preceding the shock (7,696 baht) and was substantially larger than the average monthly food consumption during this period (2,915 baht).

Are the Shocks Idiosyncratic?—Our analysis requires that the events be idiosyncratic and their occurrence be uncorrelated with trends in household behavior or network/sectoral shocks. The top panel of online Appendix Figure B4 shows that event start dates are spread evenly through the periods in the sample. Indeed, in over 87 percent of the cases, shocks affected only one household per village at any one month (bottom panel). In the bottom panel of online Appendix Table B1, we formally show that village-level trends have null predictive power on the occurrence of these events ($p = 0.39$).

Are the Shocks Exogenous?—Column 1 of online Appendix Table B1 shows within-household correlations of the probability of experiencing a shock in period $t + 1$ and contemporaneous financial characteristics, standardized with respect to the sample mean and standard deviation. We are unable to reject the null of joint significance ($p = 0.281$). In addition, the null is only rejected in one out of 13 cases at 10 percent.[12] The point estimates are also small; the largest point estimate suggests

[12] Specifically, health spending in period t is negatively correlated with the onset of the shock in $t + 1$. This is by construction: when a shock starts (i.e., when a household member starts reporting symptoms) at time $t + 1$, households are very likely to experience their largest level of health spending, hence spending in the previous month, t, will be lower.

that a 1 standard deviation increase in the value of land at t would increase the probability of experiencing a shock in $t+1$ only by 0.002. Column 2 reports p-values of the null hypothesis that the 12 lags of each household characteristic do not precede the onset of the shock (i.e., a Granger causality test). This hypothesis does not only require that households' characteristics in t do not predict the occurrence of shocks in $t+1$, but also that shocks occurring up to a year before the shock do not predict shock occurrence. We only reject this hypothesis at the 10 percent level for 1 out of the 13 variables. These patterns demonstrate that the timing of shocks is orthogonal to preshock family and business financial decisions. The next section discusses our empirical approach and addresses other potential identification concerns.

II. The Direct Effects of Idiosyncratic Shocks

To understand the indirect effects of shocks via network propagation, we first must understand how they affect the *directly* shocked household. Because the networks we study are defined by cross-household transactions of inputs, output, and labor, our first-stage analysis focuses on estimating the direct effects of shocks on business spending, labor demand, and production.

Estimating the effects of idiosyncratic shocks on household outcomes requires a valid comparison group. We would like to compare shocked households to otherwise-similar households who, by chance, were not simultaneously exposed to a shock. To implement this comparison, we follow Fadlon and Nielsen (2019) and exploit plausibly random variation in the *timing* of severe health shocks.

We compare the behavior of households who experienced a shock in period t (i.e., treated households) to that of households from the same age group and village who did not experience a shock at time t, but did experience a similar shock later on, in period $t+\Delta$ (control households).[13] Treated households are those who experienced the shock during the first half of the panel; control households experienced a shock during the second half.

We employ a difference-in-difference approach to compare changes in outcomes before and after the shock, between treatment households (who experienced an actual shock) and control households (who are assigned a placebo shock Δ periods before the occurrence of their actual shock).[14] The underlying assumption is that, in the absence of the shock, the treatment and control groups would have followed parallel trends, which we validate by testing for lack of systematic differences before the shock (parallel pretrends).

[13] Δ is determined by taking the midpoint between the months associated with the first and last shocks in each age group-village bin. Its value is approximately five years. See online Appendix B.B2 for details.

[14] Thus, if a household in the control group experiences the actual shock in t'', its placebo shock is assigned to period $t''-\Delta$. See online Appendix B.B2 for additional details.

A. *Estimation*

We estimate the following generalized difference-in-difference specification, following Fadlon and Nielsen (2019):

$$(1)\qquad y_{i,t} = \sum_{\tau=-4,\tau\neq-1}^{\tau=3} \beta_\tau \mathbf{1}\{t = \tau\} \times Treatment_i + \sum_{\tau=-4,\tau\neq-1}^{\tau=3} \theta_\tau \mathbf{1}\{t = \tau\}$$
$$+ \mathbf{X}_{i,t}\kappa + \alpha_i + \delta_t + \epsilon_{i,t},$$

where $y_{i,t}$ denotes the outcome for household i at t. Household- and month-fixed effects (α_i and δ_t) absorb time-invariant household characteristics and aggregate time-varying shocks. $Treatment_i$ is a time-invariant indicator of whether the household is in the treatment group. As each household is either observed in the treatment or comparison group, $Treatment_i$ is absorbed by the household-fixed effects. Time to treatment is denoted by $\tau_{i,t}$ and is measured in half years to increase precision. $\mathbf{X}$ is a vector of time-varying demographic characteristics including the number of male and female household members, age of the household head, and maximum years of schooling in the household. The coefficients of interest are $\{\beta_\tau\}_{\tau=-4}^{\tau=3}$, which compare differences in changes in outcomes with respect to the period immediately preceding the shock ($\tau = -1$) between households in the treatment and control groups. We focus on a two-year (i.e., four-half year) time window before and after the shock, because our panel is fully balanced during this period. We also use a more parsimonious differences-in-difference specification to compute average effects over a two-year post period:

$$(2)\qquad y_{i,t} = \beta Post_{i,t} \times Treatment_i + \theta Post_{i,t} + \mathbf{X}_{i,t}\kappa + \alpha_i + \delta_t + \epsilon_{i,t},$$

where $Post_{i,t}$ is an indicator that takes the value of 1 in periods following the shock and 0 otherwise. The parameter of interest, β, compares differences in outcomes before and after the shock between households in the treatment group and the comparison group. In both specifications, we cluster standard errors at the household level to account flexibly for serial correlation.

Our approach addresses several issues that may arise in event-study panel regressions without a stable comparison group—i.e., regressing outcomes on time- and household-fixed effects and a postshock dummy. A simple event-study approach would use all the households who did not experience a shock at period t as a control group for those who did, even those who were shocked before t. This could be problematic in our setting, because such "staggered event timing" specifications may suffer from bias when effects are heterogeneous over time (Goodman-Bacon 2018; Baker, Larcker and Wang 2022). Our design, by virtue of using a control group which is never treated, before or during the comparison window, avoids these concerns. However, this advantage comes at the cost of statistical power and limits the number of available postperiod observations as we only analyze the subset of 247 shocks that occurred earlier in the sample. Moreover, trends in outcomes may vary by age and village due to different trajectories along the life cycle; by constructing a comparison group within age group and village, our approach constructs a comparison group of households with similar preshock trends and similar risk profiles.

B. *Direct Effects: Results*

A health shock can affect households in a number of ways. Here we focus on changes in household production decisions (reducing outlays on hired labor and/or business inputs) because such dimensions are linked to cross-household transactions that determine local economic networks.

Figure 1 reports flexible difference-in-difference estimates of equation (1). Since in some cases statistical precision is limited due to our empirically demanding estimation strategy, the figure reports both 90 percent and 95 percent confidence intervals. Panel A shows that, relative to control households, shocked households experience a large and significant increase in the probability of reporting health symptoms. Panel B shows that this coincides with a sharp increase in total health expenditure.[15] Panel C shows that nonhealth consumption remains steady.

The remaining panels show that the shocks affect the household's production-side decisions. Panel D shows that, compared to households in the control group, hired labor usage declines for shocked households,[16] and panel E shows that input spending falls. Finally, panel F shows that the reduction in input spending coincides with a reduction in revenues. The graphical evidence also documents parallel pretrends for all six outcomes.[17]

The onset of the declines in input spending and revenues coincide with the sharp increase in health spending induced by the shock in the half-year of the shock's occurrence. Over this same period, we observe a smooth trajectory of nonhealth spending. This suggests that shocked households meet short-term liquidity needs in part by drawing down working capital. The effects on input spending and revenues also persist over time. This may partly reflect the suspension of key activities in the household due to the shock, as shown in panel F of online Appendix Figure A2. A short-term health spending need can also generate persistent effects on business spending through other channels. For instance, households may relinquish their fixed assets, inventories or livestock, inducing a persistent reduction of a business's scale, especially when they exhaust other sources of liquidity. By event period 2, household assets appear to decline (see online Appendix Figure A5b), although the declines are not estimated with precision.[18] Likewise, shocked firms may not

[15] Online Appendix Figure A2 provides evidence that the onset of the shock coincides with changes in health status and health spending that are likely severe and unexpected. Panels A–E show that the usage of and spending on outpatient and inpatient care increase and that the probability of receiving medical care due to an accident also increases. These patterns suggest that the shocks generate immediate household spending needs that squeeze a household's budget. The nonfinancial consequences of the shocks may be persistent: panel F of online Appendix Figure A2 shows that the shocks increased the probability of suspending activities for more than a week and that even though this effect declines over time, it persists for two years after the onset of the shock.

[16] One concern with Figure 1, panel D is that at event period -4, hired labor appears higher for treated households, relative to those experiencing a placebo shock. Online Appendix Figure A3 reports results including one extra half year in the preperiod. The pretrend at event time -4 appears to be a one-off deviation, ruling out the concern that systematic pretrends drive the results.

[17] Online Appendix Figure A4 shows the same dynamics in the raw data.

[18] Column 3 in panel A on online Appendix Table A15, finds a large though not significant decline in total noncash assets (fixed assets, livestock and inventories).

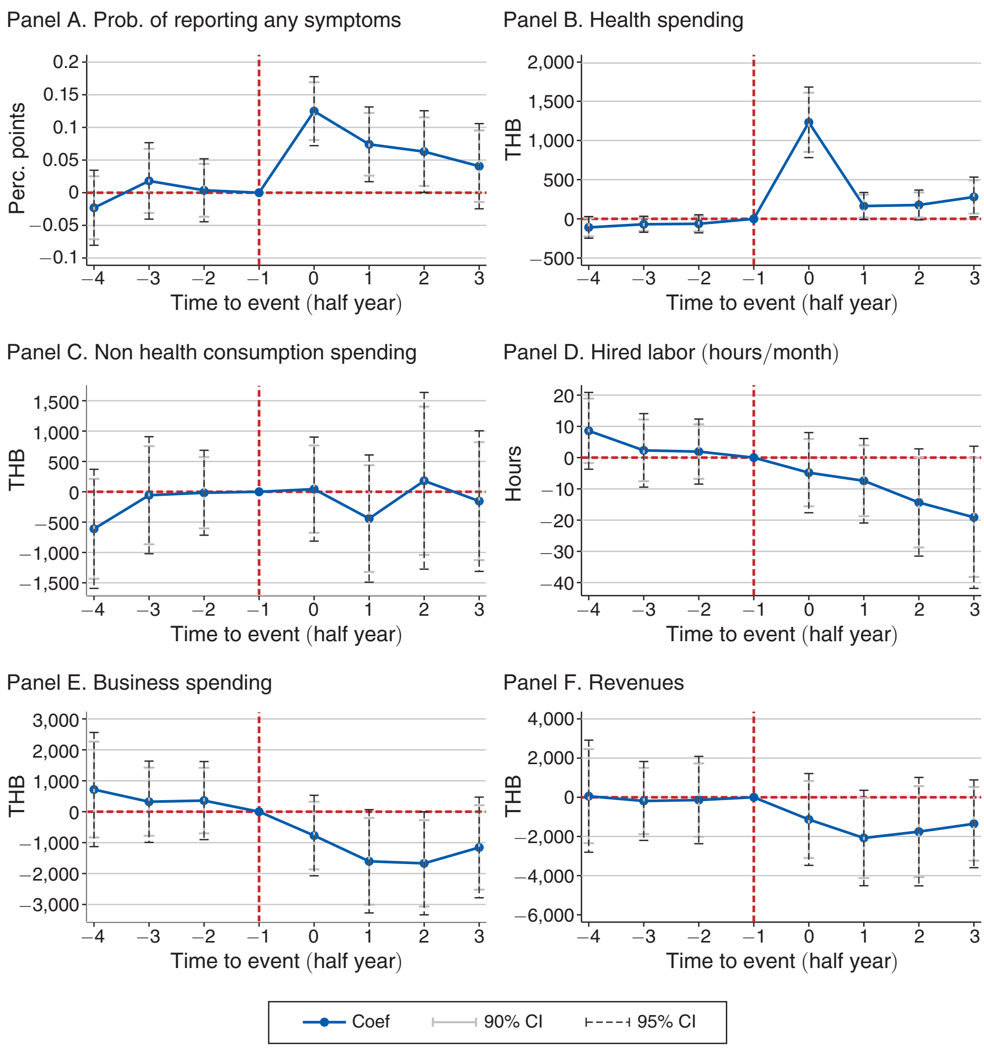

Figure 1. Direct Effects of Health Shocks

Notes: Each dot represents differences between treatment and control households in changes in outcomes relative to the period preceding the beginning of the shock ($\tau = -1$). The estimating sample includes two years before and after the shock divided in half-year bins. All specifications control for household time-variant demographic characteristics, as well as household and month fixed effects. 90 percent and 95 percent confidence intervals are computed using standard errors clustered at the household level. Costs and revenues exclude costs and earnings associated with the provision of labor to other households or firms. All variables measured in Thai baht are winsorized with respect to the 99 percent percentile.

necessarily face the same demand for their products after the shock if consumers deepen their links with other providers. Both channels (reduction in business investment and suspension of activities) suggest that, unlike the case of large firms owned by a diversified set of investors, the business outcomes of small firms owned by un-hedged households respond to health shocks affecting household endowments; a behavior that is inconsistent with the separation theorem (Benjamin 1992).

To provide a quantitative assessment of the overall impact of the health shocks, in Table 1 we report difference-in-difference estimates of the effect of the shock

Table 1—Effects on Health, Spending, and Family Businesses

	Reported symptoms (1)	Health spending (2)	Nonhealth spending (3)	Total spending (4)	Business spending (5)	Hired labor (6)	Household labor (7)	Revenues (8)
Panel A. Using shocks occurring during the first half of the sample								
Post × Treatment	0.0764	530.5	77.56	608.0	−1,642.2	−14.49	−11.63	−1,514.6
	(0.0229)	(90.29)	(335.3)	(358.8)	(793.5)	(7.648)	(8.589)	(1,011.3)
Baseline mean (DV)	0.361	151.8	5,260.4	5,412.2	7,307.5	17.58	151.7	14,304.5
Observations	23,014	23,015	23,015	23,015	23,015	23,015	23,015	23,015
Number of events	249	249	249	249	249	249	249	249
Adj. R^2	0.232	0.0487	0.148	0.155	0.789	0.600	0.713	0.641
Panel B. Using all shocks								
Post × Treatment	0.0839	411.9	234.5	646.4	−1,325.3	−9.918	−15.05	−1,771.9
	(0.0167)	(61.90)	(331.2)	(336.8)	(515.2)	(4.885)	(6.457)	(664.4)
Baseline mean (DV)	0.345	158.2	5,770.4	5,928.6	7,172.5	15.81	140.3	14,302.3
Observations	43,923	43,925	43,925	43,925	43,925	43,925	43,925	43,925
Number of events	476	476	476	476	476	476	476	476
Adj. R^2	0.226	0.0443	0.0993	0.108	0.758	0.685	0.657	0.570

Notes: The table reports estimates of β from equation (2) for different outcomes. Each column reports differences between treatment and control households in changes in outcomes before and after the shock. All regressions control for household demographic characteristics, household and month fixed effects. Standard errors are clustered at the household level. Business spending, labor, and revenues are aggregated across all businesses operated by household members, and exclude revenues and costs of wage labor provision to other businesses or households. Hired labor and labor provided by household members are measured in hours/month.

on outcomes over a 24-month postshock period, corresponding to equation (2). Panel A restricts the treatment sample to shocks occurring in the first half of the period, ensuring the control group is never treated before or during the comparison window. Column 1 shows that the shocks are associated with a significant increase in the likelihood of reporting a health issue and column 2 shows that the shock leads to a large increase in health spending. While this is by construction, the magnitude, approximately 530 baht, is notable, representing a roughly 350 percent increase relative to the baseline mean (151.8 baht). Column 4 shows that during the two years following the shock, total spending increases for shocked households, relative to control households, by approximately 608 baht on average, an amount close to the effect on health spending. Thus, in terms of nonhealth spending (column 3), shocked households appear to fully buffer the shocks.

Buffering consumption may entail costly adjustments by shocked households (Chetty and Looney 2006). Indeed, affected households significantly decrease spending on business inputs (column 5) and reduce the use of external labor (column 6).[19,20] Households also appear to reduce the use of labor provided by household members (column 7), though the effect is not significant when we restrict to shocks occurring in the first half of the period. As a result of these reduced

[19] Households may also engage in other strategies to cope with the shocks, such as receiving gifts, borrowing, relinquishing fixed and liquid assets, and using unpaid external labor. See online Appendix B.B5. We discuss the effects on gifts in more detail in Section IVC.

[20] An alternative mechanism is that nonshocked household members reduced the time allocated to the operation of family businesses in order to take care of the shocked household member. However, online Appendix Table A2 shows that the overall household-level number of days dedicated to housework appears to decline as well.

investments in inputs and labor, there is a decrease in the revenues from family enterprises, as seen in column 8, albeit imprecisely estimated (p-value $= 0.134$).[21]

Panel B reports results that also include early-shocked households as controls for late-shocked households,[22] which roughly doubles the number of events. Reassuringly, the estimates are very similar to those in panel A but are estimated with more precision.[23]

Robustness.—Panel A of online Appendix Table A3 shows robustness to alternative definitions of health shocks.[24] Columns 1 and 2 show that using the largest *change* in health spending throughout the panel to identify episodes of high health spending yields very similar results to the main specification, which is based on the highest spending *levels*.[25] One concern is that our approach may include shocks that, based on their magnitude, are innocuous, despite being the largest shocks experienced by the households throughout the panel. In columns 3 and 4, we exclude shocks associated with expenditure levels that fall in the bottom 75 percent of the postshock health spending distribution for control households; the results are similar to those of our main specification. We also employ two alternative ways of selecting shocks: a household-specific benchmark (health spending larger than the preshock average food consumption) and a common benchmark (health spending larger than the sample mean plus one standard deviation). Columns 5 to 8 show that we obtain qualitatively similar results. Both approaches select larger shocks and thus yield larger effect sizes, but also use a smaller set of shocks, reducing statistical power.[26]

In columns 3 and 4 in online Appendix Table A4, we define shocks based on whether a household member suspended their main activities for at least one week. We use a one-week threshold based on Gertler and Gruber (2002), who show that only severe health shocks yield effects on household spending. As opposed to our main specification, which by construction captures episodes of financial stress with potential effects on time allocation, this alternative definition captures shocks to time availability with potential financial implications. Reassuringly, the results are qualitatively similar.[27] The results are robust to allowing for multiple, nonoverlapping shocks per household, as shown in online Appendix Table A5.

[21] The point estimates of the effects on business spending and revenues are quite similar at approximately 1,642 baht and 1,514 baht, respectively, suggesting that business profits are unaffected. This can be a consequence of the reduced business scale induced by the decline in business spending. The zero effect on profits may also reflect households responding to the reduction in earnings by taking costly actions that in the short run reduce costs but are harmful in the long term (e.g., deferring needed maintenance of equipment).

[22] The treatment status of control households is held fixed around the 24-month analysis window around each event. This addresses potential biases in difference-in-difference frameworks that can arise when treatment status varies over time.

[23] The effects on household labor and revenues are now significant at conventional levels, and all other outcomes remain significant.

[24] See online Appendix Section B.B2 for a detailed description of each shock definition.

[25] This similarity reflects the fact that while we use spending data to identify the episodes (either maximum levels or changes), we use symptoms data to calculate the onset of the shocks; see footnote 11 and online Appendix B.

[26] In online Appendix Table A5 we show that power is increased in these specifications when we include multiple, nonoverlapping shocks per household (columns 3 and 4). However, doing so comes at the cost of imposing two additional identification assumptions. First, shocks experienced earlier on should not affect the probability of experiencing another health shock in the future and second, the effects of earlier shocks should not have long-lasting effects on the trajectories of outcomes that can lead to violations of the parallel trends assumption.

[27] Note that these effects are less precisely estimated, because this alternative definition identifies fewer shocks. We provide a brief discussion and robustness to alternative definitions in online Appendix Section B.B2 and online Appendix Table A4.

The results are also robust to using alternative control groups (see online Appendix B.B2 for details). Online Appendix Table A6 columns 1 and 2 show that the results are robust to randomly allocating the placebo shocks. Columns 3 and 4 use all not-yet-shocked households in the same village as controls to shocked households at time t using a stacked difference-in-difference specification. This specification, which uses the not-yet-treated as the control group, is recommended in the recent difference-in-difference literature (Baker, Larcker, and Wang 2022). Columns 5 and 6 show robustness to using not-*currently* shocked households as controls in a standard two-way fixed effects specification, and columns 7 and 8 report point estimates using Callaway and Sant'Anna's (2021) estimator with households treated in the second half of the sample as controls.[28] Reassuringly, results using these specifications are consistent with the main specification. Likewise, online Appendix Table A7 shows that the effects are unchanged when using an unbalanced panel of households for the main specification.

Finally, online Appendix Table A8 presents co-movements between health status and spending using all households in the sample and all survey periods. As expected, changes in health status coincide with changes in health spending. Moreover, changes in health status driven by uncommon health conditions (those more likely to coincide with the spending shocks that we study) predict larger changes in health spending and declines in business spending.

III. Economic Networks and the Propagation of Shocks

The results above show that health shocks meet the necessary criteria for understanding propagation: their timing is exogenous, their occurrence is idiosyncratic, and the shocks have substantial effects on household production decisions. Given the significant degree of interlinkage in the study villages, we next examine whether these shocks propagate to other households. We analyze two propagation channels. First, shocks could propagate through local supply chain networks: health shocks lead to decreases in the supply and demand for inputs, which could lead to reductions in sales and revenue for those households who trade with shocked households. Second, shocks could propagate through local labor networks: as supply and demand for outside labor decrease due to the shocks, households who exchange labor with shocked households could suffer falls in hours, earnings, and revenue.

A. *Identifying Propagation Effects*

We exploit two sources of variation to test whether idiosyncratic health shocks propagated to other agents in the local economy. First, we use variation in the timing of each household-level shock. Second, we use the fact that a household's exposure will depend on their network connections to the shocked household, via the supply chain or labor market networks, or both. We assess the propagation of idiosyncratic shocks to other local family businesses by comparing households who, before the shock, shared closer market linkages with household j's businesses to those who

[28] The corresponding event-study estimates are reported in online Appendix Figure A6.

were not or less-connected to household *j* before the shock, before versus after the shock to household *j*.

Throughout our sample period, we observe multiple health shocks per village. We construct a dataset capturing information of nonshocked households before and after each health shock in the sample. For each event, we take two years of pre- and postshock observations of households living in the same village as the directly shocked household.[29] We then stack the observations into a dataset at the household (i) by time (t) by event (j) level for each village.

We combine this dataset with information on network connections between the shocked household (j) and other households (i) in the village, measured during the year preceding the shock to household *j*. We use preshock networks as links may respond to economic shocks themselves. The assumption is that households who transacted with the shocked household during the preperiod, on average, would have been more likely to transact with the shocked households in the postperiod, in the absence of the shock.[30]

We measure exposure as the inverse distance in the undirected village network between household *i* and the shocked household *j*: $Closeness_{i,j} = 1/dist_{i,j}$.[31] As households are further away in the network from shocked households, exposure (closeness) decreases. We begin by computing overall closeness based on transactions in the supply chain or labor market networks, because households can be exposed through either network. To distinguish between exposure in the supply chain and labor market networks, we also compute measures of closeness in each separate network (see Section IIIC).

We elicit economic networks using survey instead of census data (Chandrasekhar and Lewis 2016). Thus, it is possible that we underestimate the closeness of some sample households to shocked households.[32] Because we may be underestimating exposure—classifying some households as not or less-exposed when they are actually (more) exposed—our results could be biased toward zero. Thus, we interpret our magnitudes as *lower bounds* of the indirect effects of idiosyncratic shocks on other households.

Not all shocked households are active in the local markets for goods and not all shocked households employ or work for other villagers. Thus, we analyze the propagation of shocks through village networks by focusing only on events corresponding

[29] We restrict the analysis to two years before and after the shock to be consistent with the analysis of the direct effects of the shocks and because we only have a fully balanced panel for this time window.

[30] This is consistent with the evidence of persistence in the village networks discussed in Section IIIC and with evidence of the importance of time-invariant determinants of economic connections, such as kinship relations (Kinnan and Townsend 2012), race or caste (Munshi 2014), and the existence of economic frictions such as contracting issues that may limit trade between households (Ahlin and Townsend 2007) or between firms (Aaronson et al. 2004) in local economic networks.

[31] This measure equals 1 if household *i* has directly traded with the shocked household *j* and 0 if household *i* does not have any direct or indirect connections with the shocked household. The geodesic distance between two unconnected nodes is $dist_{i,j} = \infty$ and so their closeness equals 0 in that case. We focus on undirected networks because the shock can propagate both up- and downstream, as we document in Section IIIB. By undirected networks we mean that we do not distinguish between incoming versus outgoing transactions. Likewise, we weight each transaction equally for our calculations.

[32] See footnote 8 for a discussion of this issue.

to the 410 households who traded in either the supply chain or labor market networks during the year preceding their shock out of the 476 shocks in our sample.[33]

We estimate the following generalized difference-in-difference specification:

$$(3)\quad y_{i,t,j} = \sum_{\tau=-4,\tau\neq-1}^{\tau=4} \beta_\tau \mathbf{1}\{t = \tau\} \times Closeness_{i,j} + \gamma Closeness_{i,j} + \mathbf{X}_{i,t,j}\kappa + \alpha_i + \omega_j + \delta_t + \theta_{\tau(j)} + \delta_t \times Degree_i + \epsilon_{i,t,j},$$

and the parsimonious difference-in-difference specification:

$$(4)\quad y_{i,t,j} = \beta Post_{t,j} \times Closeness_{i,j} + \gamma Closeness_{i,j} + \mathbf{X}_{i,t,j}\kappa + \alpha_i + \omega_j + \delta_t + \theta_{\tau(j)} + \delta_t \times Degree_i + \epsilon_{i,t,j},$$

where y denotes the outcome of interest for household i in village v at time t around the shock suffered by household j. In the "event-study" specification (equation (3)), τ denotes a half year, which may precede ($\tau < 0$) or follow ($\tau \geq 0$) the shock to household j. $Closeness_{i,j}$ denotes inverse distance to the shocked household during the year preceding the shock to j.[34] The coefficients of interest in equation (3) are β_τ, which capture relative changes in outcomes corresponding to half year τ with respect to the half year preceding the event ($\tau = -1$) associated with one additional unit of closeness (i.e., between more- versus less-exposed households). In the generalized difference-in-difference specification, equation (4), $Post_{t,j}$ takes the value of 1 during the two years following the shock to household j and 0 for the preperiod. The coefficient of interest, β, captures differences in outcomes associated with one additional unit of closeness, with respect to the preperiod.

Controls include household fixed effects (α_i); month fixed effects (δ_t); shocked-household fixed effects (ω_j); time-to-shock fixed effects ($\theta_{\tau(j)}$), which account for village-specific time-varying shocks during the analysis window corresponding to the shock to household j; and a vector of time-varying demographic characteristics ($\mathbf{X}_{i,t,j}$).[35] We also control for time-varying trends for more-central households, who could also be more likely to be close to other households, by including interactions of the number of links of household i ($Degree_i$) during the year preceding the shock to j with time fixed effects. Thus, we are in essence comparing two households equally well-connected to the network, one of whom happens to be closer to the shocked household. We use two-way clustered standard errors at the event level j and household level i to allow for flexible correlation across households during the periods preceding and following event j and across responses of the same household i to different events. In order to focus on indirect effects, we drop observations of directly shocked households (where $i = j$) from the analysis and

[33] In online Appendix Table A10 we report similar results when we also use shocks to unconnected households (columns 3 and 4), coding closeness equal to zero for all nonshocked households.

[34] Below we consider several definitions of *Closeness*: proximity in the overall network pooling the supply chain and labor market networks, as well as proximity in either network.

[35] These are household size, gender composition, average age, and years of schooling.

exclude observations of households who experienced their own shock within a year before or after the shock to household *j*.

The identifying assumption underlying our strategy for estimating indirect effects is that, in the absence of the shock to household *j*, the outcomes of households *i* and i', with differential closeness to *j*, would have evolved following parallel trends, conditional on the vector of controls included in equations (3) and (4). We validate this by testing for (lack of) differences in the preperiod: for $\tau < 0$, we verify that β_τ is not different from 0.[36]

B. *Results: Propagation through Economic Networks*

Figure 2 presents flexible difference-in-difference estimates following equation (3).[37] Panel A analyzes total transactions. After a health shock, households who are more connected to shocked households differentially reduce the number of transactions with other households in the village. Prior to the shock, transactions are not different for closer versus more-distant households. After the shock, however, transactions decline more for households who are closer to the shocked household. Panels B and C show that supply chain and labor-network transactions, respectively, each exhibit the same pattern seen for total transactions. Panel D shows that, as local networks are shocked, total income declines for households closer to the shocked household. In all four cases, the preshock period shows no evidence of differential pretrends. Finally, Panel E shows an analogous result for total consumption expenditure, which declines in the postshock period (and exhibits no differential trend in the preperiod).

The effects on transactions, income, and spending are evident in all three half-year periods following the shock and do not appear to shrink in magnitude over time: the effects are quite persistent. In theory, indirectly, hit households might attenuate these effects over time by finding new local trading partners. However, the evidence on the rigidity of local networks shown below (Section IIIC) demonstrates that such reorganization of local ties is very difficult, at least over the span of one to two years.

Table 2 panel A shows difference-in-difference estimates corresponding to equation (4). It documents significant postshock declines in the number of monthly transactions in the supply chain (column 1) and labor market networks (column 2), and in total transactions (column 3). These effects are large, representing declines of 19 percent, 24 percent, and 21 percent relative to the preperiod means, respectively. Column 4 shows that these changes in turn reduce income: a one-unit increase in *Closeness* is associated with a fall in income of 820 baht, or 8 percent of the preperiod mean. In turn, consumption spending falls by 207 baht, or 2.7 percent of its preperiod mean (column 5).[38] The fall in consumption is smaller than the fall in income, suggesting that indirectly shocked households can partly, but not

[36] As discussed below, panel B of Table 2 shows that proximity in the labor market does not predict changes in supply chain transactions and vice versa is additional evidence in support of this assumption.

[37] Since in some cases statistical precision is limited due to our empirically demanding estimation strategy, the figure reports both 90 percent and 95 percent confidence intervals.

[38] Recall that these are effects associated with moving from *Closeness* $= 0$ (unconnected to the directly shocked household) to *Closeness* $= 1$ (directly linked). The mean level of *Closeness* $= 0.42$ (see Table 4), so the average indirect effect is 42 percent of the coefficient.

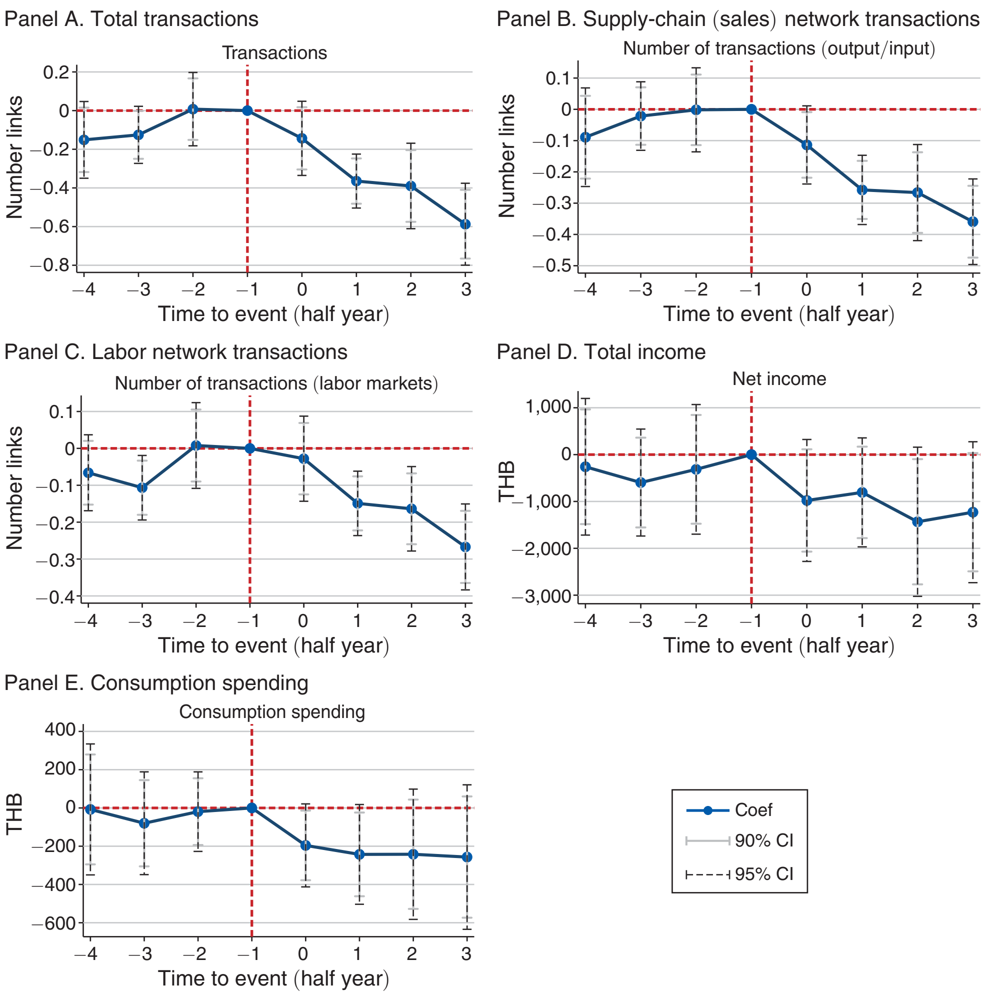

Figure 2. Indirect Effects on Transactions, Income, and Consumption

Notes: The figure presents flexible difference-in-difference estimates of the indirect effects of idiosyncratic shocks on local businesses, following equation (3). All regressions include household fixed effects, event fixed effects, month fixed effects, village- and year-fixed effects, and household size, household average age and education, and the number of adult males and females in each household. Each dot captures differences in changes in outcomes with respect to the half-year preceding the shock (-1) between more- and less-exposed households. Standard errors are two-way clustered at the household (i) and shock level (j). All variables measured in Thai baht are winsorized with respect to the 99 percent percentile.

completely, smooth their indirect shock exposure. Moreover, the consumption fall increases over time, suggesting the depletion over time of consumption-smoothing strategies (e.g., savings).[39]

[39] Why are directly shocked households able to smooth their consumption, while *indirectly* hit households are not? One explanation is that directly shocked households see economically and statistically significant increases in transfers (see online Appendix Figure A5a), while indirectly shocked households do not (see online Appendix Table A12). This difference in turn may be due to the fact that the direct shocks are large increases in health spending, often associated with changes in health symptoms. These shocks are salient and relatively observable. The

Table 2—Propagation of Idiosyncratic Shocks

	Input/output (1)	Hired labor (2)	All transactions (3)	Total income (4)	Total spending (5)
Panel A. Propagation effects through village networks					
Post × *Closeness*	−0.19	−0.11	−0.30	−820	−207
	(0.06)	(0.04)	(0.08)	(412)	(151)
Observations	434,145	434,145	434,145	434,145	434,145
R^2	0.44	0.23	0.37	0.21	0.64
Preperiod mean	0.989	0.462	1.451	10,729	7,433
Number of events	410	410	410	410	410
Panel B. Propagation effects through supply chain and labor market networks.					
Post × *closeness* (supply chain network)	−0.24	0.02	−0.22	106	155
	(0.06)	(0.04)	(0.08)	(464)	(174)
Post × *closeness* (labor market network)	−0.02	−0.20	−0.22	−1,233	−501
	(0.06)	(0.04)	(0.08)	(423)	(153)
Observations	434,145	434,145	434,145	434,145	434,145
R^2	0.44	0.23	0.37	0.21	0.64
Preperiod mean	0.989	0.462	1.451	10,729	7,433
Number of events	410	410	410	410	410

Notes: Panel A presents estimates of β from equation (4). $Closeness_{i,j}$ denotes inverse distance to the shocked household during the year preceding the shock to j. Each coefficient captures differences in changes in outcomes before and after the shock between more- and less-exposed households, through village networks. Each regression includes household (i), event j, and month fixed effects, as well as demographic characteristics such as household size, average age, education, and number of male and female adults. Panel B presents estimates of β from equation a variation of (4) where $Closeness_{i,j}$ denotes inverse distance to the shocked household during the year preceding the shock to j, by type of network. Standard errors are two-way clustered at the household (i) and event (j) level.

C. *Mechanisms*

Effects of Supply Chain versus Labor Market Exposure.—In Table 2 Panel B, we examine whether the effect of exposure through the supply chain network has different effects than exposure through the labor market network. If proximity through the supply chain (labor) network is associated with changes in input/output (hired labor) transactions, and not vice versa, this is supportive of the identification assumption, because many plausible confounds (e.g., differential trends between closer versus more-distant households) would manifest in both sets of outcomes. Because the two networks are correlated, we analyze the effect of preshock exposure to one controlling for preshock exposure to the other.[40] Column 1 shows that, conditional on proximity in the labor market network, a 1-unit increase in proximity in the supply chain network is associated with a significant fall in input/output transactions of 0.24. There is no effect on input/output transactions associated with proximity through the labor market network. Analogously, column 2 shows that

indirect shocks, on the other hand, arise from reductions in supply and demand facing household businesses. Such shocks are likely less salient and potentially more subject to concerns of effort and verifiability, hence potentially less insurable. Moreover, because the indirect shock, by its nature, affects many interlinked households, the shock becomes de facto aggregate, which makes the potential for insurance via gifts from other villagers more limited. We return to this issue in online Appendix B.B5.

[40] On average, 43 percent of households share a direct or indirect link to the shocked households through both networks, 17 percent are directly or indirectly linked to the shocked household through only the supply chain network, 13 percent are directly or indirectly connected to the shocked households through only the labor market network and 27 percent of households are not connected to the shocked households either network.

proximity through the labor market network has a negative and significant effect (-0.20) on transactions involving paid labor, while there is no effect seen via the supply chain network. In column 3, proximity via the supply chain network or the labor market network generates negative and significant effects on the total number of transactions of -0.22 in both cases.

Columns 4 and 5 show that proximity via the labor market network is associated with large and significant drops in both income and consumption, while the corresponding effects of proximity via the supply chain network are small and insignificant. Thus, while shocks propagate through both networks, the severity of the impacts are larger when shocks transmit through labor market networks. This result underscores the importance of distinguishing between networks and suggests estimates that consider supply chain networks alone may be lower bounds.

Why Do We Observe Greater Propagation via Labor Networks?: A possible explanation is that, although the absolute effect of exposure via supply chain networks on input/output transactions is similar to the effect through labor market networks on labor transactions, the effect on labor market transactions is larger in relative terms. When shocks propagate through the labor market, labor market transactions fall by 43 percent relative to the pre-period level, while, when the shock propagates through the supply chain, the decline in transactions of inputs and final goods represents 23 percent of the pre-period mean (see columns 1 an 2 in panel B of Table 2). It may also be more difficult for households to adjust along the intensive margin: in the goods market, fewer but larger within-village transactions or more out-of-village transactions may substitute for the loss of transactions with the shocked household. In the labor market, working more hours for other employers or forming new employment connections may be more difficult, due to concerns of adverse selection (Barwick et al. 2019), finite time in the day, travel costs, etc. In addition, as we discuss below, shocks reverberate through the networks, reducing the demand for labor at the local level and the local availability of jobs. Selling labor in other villages can be difficult, as temporary migration can be costly (Bryan, Chowdhury, and Mobarak 2014).

The Ripple Effects of Shocks

Indirectly Connected Households: The declines in overall transactions that we observe are strongest for directly connected households—those one link from the shocked households—but affect indirectly connected households as well (see Figure 3).[41] When, due to a shock, those linked directly to shocked households reduce sales of goods or labor (outgoing transactions to the shocked household), this leads to declines in income and consumption, though less-precisely estimated. As households are consumers but also operate firms, these indirect effects translate

[41] Figure 3 plots indirect effects decomposing the measure of closeness into four categories: directly connected households (I link away from the shocked households), households who are two or three links away from the shocked households, those who are four or five links away from the shocked households, and the base category, those who are six or more links away in the network, including those who are unconnected to the shocked household. Although the effects dissipate through the network, there are nonnegligible propagation effects on indirectly connected households.

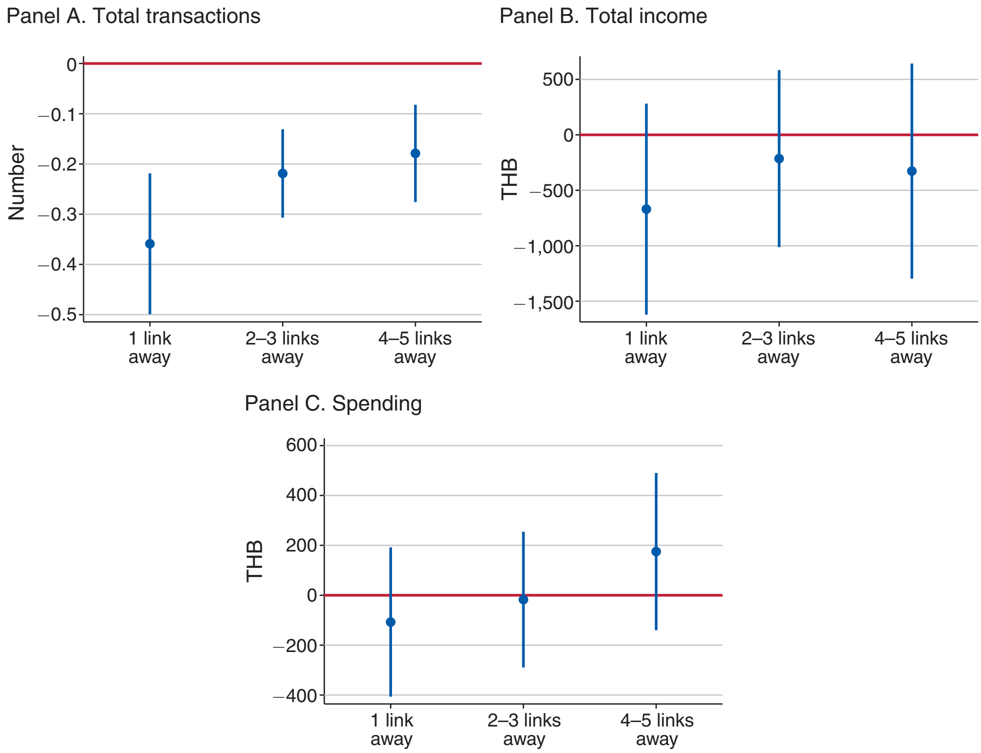

Figure 3. Indirect Effects by Distance to Shocked Households

Notes: The figure depicts indirect effects of the shocks based on distance to the shocked household in the preshock network. The coefficients correspond to a regression of the dependent variable on a postshock indicator, distance-to-shocked household dummies, and interactions of the postshock indicator and the distance dummies. The base distance category is households that are more than five links away from the shocked households or that are unconnected to the shocked household. All regressions include household fixed effects, event fixed effects, month fixed effects, household size, household average age and education, the number of adult males and females in each household, and control for degree centrality interacted with month fixed effects. 95 percent confidence intervals are based on standard errors that are two-way clustered at the household (i) and shock level (j). All variables measured in Thai baht are winsorized with respect to the 99 percent percentile.

into fewer purchases (incoming transactions) of goods, inputs and labor from other households, triggering further propagation through the network. As a result, what was initially a negative shock to one household turns into a negative aggregate demand shock.

Upstream versus Downstream Propagation: Table 3 shows that the fall in overall transactions documented above is driven by falls in both outgoing transactions (sales of inputs and labor) and incoming transactions (purchases/hiring). In sum, the health shocks that we study generate indirect effects both upstream and downstream, as the costly adjustments taken by the directly shocked household reverberate through local networks. Shocks that are prima facie idiosyncratic are spread to other connected households.

Dynamics and Persistence.—The indirect effects that we document appear to persist over time, despite the transitory increase in health spending due to the health

Table 3—Propagation Effects on Outgoing and Incoming Transactions

	Input/output		Labor		All transactions	
	Outgoing (1)	Incoming (2)	Outgoing (3)	Incoming (4)	Outgoing (5)	Incoming (6)
Post × *Closeness*	−0.07	−0.12	−0.08	−0.03	−0.16	−0.15
	(0.04)	(0.03)	(0.02)	(0.03)	(0.05)	(0.04)
Observations	434,145	434,145	434,145	434,145	434,145	434,145
R^2	0.53	0.26	0.15	0.21	0.43	0.25
Preperiod mean	0.495	0.494	0.178	0.284	0.673	0.778
Number of events	410	410	410	410	410	410

Notes: The table presents estimates of β from equation (4). $Closeness_{i,j}$ denotes inverse distance to the shocked household during the year preceding the shock to j. Each coefficient captures differences in changes in outcomes before and after the shock between more- and less-exposed households, through village networks. Each regression includes household (i), event j, month fixed effects, and demographic characteristics such as household size, average age, education and number of male and female adults. Standard errors are two-way clustered at the household (i) and event (j) level.

shocks. This is also true when we expand the postshock analysis window as in online Appendix Figure A7.[42] The existence of relatively persistent declines in business activity among directly shocked households, the inability of households to immediately replace broken links, and the reduced demand for labor at the village level may explain the highly persistent effects on indirectly shocked households.

While the direct effect on health spending peaks within a half-year of the changes in health status, there is evidence of longer term health consequences (see Figure 1, panels A and B). In addition, directly shocked households reduced business spending and relinquished business assets (see Figure 1, panel E and online Appendix Figure A5b), reducing the business scale. To the extent that replacing such assets takes time, the reduction in business scale may have generated a persistent decrease in labor demand, which could explain the persistent *indirect* effects.

In addition, as directly shocked households reduce their demand for workers and physical inputs, indirectly shocked households may struggle to replace the broken links. Thus, frictions in rewiring economic networks may lead to a large degree of persistence. To document the degree of rigidity in the local networks, we construct a dyadic dataset that includes indicators of whether each pair of sample households (dyads) transacted in year t either in the local goods, labor, or financial market and estimate the extent to which past transactions predict future transactions, conditional on measures of similarity and connections based on kinship networks at baseline. (See online Appendix B.B4 for details.) Online Appendix Table A9 shows that labor market and supply chain networks exhibit a striking degree of rigidity over time. Column 4 in panel B shows that dyads linked through the labor market network at period $t-1$ are 33 percentage points more likely to transact in period t, relative to unconnected dyads at $t-1$. This level of persistence is an order of magnitude above

[42] Note, however, that a causal interpretation of the results based on a larger analysis window impose additional assumptions. By construction, none of the nonshocked households at event period $\tau = 0$ suffered a direct shock within two years. Thus, a longer-term analysis as in Figure 7 requires assuming that no other household suffered a shock during the post shock period. This assumption is less plausible for longer analysis windows.

the probability that two randomly chosen nodes in the network transact in a given year in the labor market (0.0612) or supply chain (0.0508); the persistence in the supply chain and labor market networks is also greater than that seen in the gift and loan network (panel C).[43]

Finally, as discussed in Section IIIC, as the direct effects of the shocks ripple through the networks, the local aggregate demand may have declined. This decline in aggregate demand may have stronger consequences in local labor markets. To the extent that replacing local employers with external employers is costly, the decline in local economic activity may generate these persistent impacts.

D. *Robustness*

The indirect effects are robust to a battery of alternative specifications. Online Appendix Table A10 shows results controlling for village-month fixed effects (columns 1 and 2), including shocks to unconnected households during the pre-period (columns 3 and 4), using an unbalanced panel of households to estimate indirect effects (columns 5 and 6), and excluding shocks to large firms to attenuate issues of granularity as in Gabaix (2011) (columns 7 and 8).[44] The results are also robust to utilizing alternative identification strategies (see online Appendix Section B.B3 for details). Columns 1 and 2 of online Appendix Table A11 report estimates from a triple-difference specification using the placebo shocks from Section IIA as a control group.[45] Columns 3 and 4 show results from an alternative identification strategy that parallels our strategy for estimating the direct effects. In this alternative strategy, we compare households who are indirectly shocked for the first time at time t to a control group of households who will be indirectly exposed to a shock for the first time several years in the future. The results are remarkably similar to those from our main specification. Finally, panel B of online Appendix Table A3 shows that our main effects are qualitatively similar, albeit less precisely estimated, when we use alternative definitions of shocks.[46]

IV. The Aggregate Effects of Idiosyncratic Shocks

So far, our analysis shows that idiosyncratic shocks propagate through economic networks, and that shocks spread through labor market networks appear to have larger negative indirect effects on income and consumption. A natural question is whether these microeconomic effects can have important aggregate implications, and, if so, which features of local networks contribute to the amplification or attenuation of these shocks. In this section, we provide evidence to answer these questions.

[43] Our results on persistence in labor market networks echo Felix (2022), who documents highly inelastic within-market cross-firm substitution in Brazil.

[44] We drop shocks to firms with revenues (over the 12 months preceding the shock) that are above the median revenues among shocked firms.

[45] In this case, we append data on households with different degrees of closeness to placebo shocks to our dataset on indirectly shocked households and fully interact equation (4) with a Treatment/Placebo dummy. Columns 9 and 10 of online Appendix Table A10 report the coefficient on the triple interaction (*Post* $\times$ *Closeness* $\times$ *Treatment*).

[46] This is largely due to the fact that more-stringent definitions of shock identify fewer shocks.

Table 4—Multiplier Effects

	Median		
Panel A. Preshock characteristics			
Market share (shocked household)		0.028	
Average closeness (shocked household)		0.422	
Number of indirectly exposed households (preshock)		23	
	Point estimate		
Panel B. Treatment effects			
Direct effects on business spending (baht)		−1,642	
Direct effects on business spending (% relative to preshock mean)		−22.5	
Indirect effects on consumption spending (baht)		−207	
Indirect effects on consumption spending at mean closeness (baht)		−88	
Indirect effects on consumption spending (Percent of preshock means)		−1.16	
	Estimate %	95 CI	90 CI
Panel C. Aggregate effects			
Multiplier	1.23	[−0.394, 6.025]	[−0.092, 4.032]

Notes: Panel A reports medians across shocks. Median market share is computed as the ratio of a shocked household's total revenues during the year preceding its shock (excluding labor income) divided by village aggregate value added measured during the same period as in (Hulten 1978). Avg. Closeness and the number of indirectly exposed households are computed across all nonshocked households in the same village of a shocked household. Households who suffer a direct shock themselves within a year of the indirect shock are excluded from the calculations. Panel B reports direct and indirect treatment effects based on column 5 of Table 1 and column 5 in Table 2. Panel C reports back-of-the-envelope calculations. Confidence intervals are based on percentiles of 500 bootstrap replications.

A. *The Multiplier Effect of Idiosyncratic Shocks*

What is the total magnitude of indirect effects relative to that of direct effects? The former are larger on a per-household basis, but the latter can potentially affect many more households. In order to compare their overall magnitudes, and so obtain an estimate of the overall "multiplier effect" of the fall in business spending associated with the shock, we perform a back-of-the-envelope exercise to estimate the total indirect fall in consumption for each baht of reduced business spending by directly affected households.

Table 4 summarizes the key values. The indirect effect on consumption associated with a 1-unit change in *Closeness*, from column 5 in panel A in Table 2, is a fall of −207 baht. The median level of *Closeness* in the village network is 0.42 and the median number of indirectly exposed households (i.e., households who are connected to the shocked household via the network) is 23.[47] The implied total indirect effect using mean values is $-207 \times 0.42 \times 23 = -1{,}999.62$ baht per month.

From column 5 in panel A in Table 1, the fall in business costs for a directly affected household is −1,642 baht, so the indirect effects using median closeness represent a multiplier effect of 1.23 (column 1, panel C in Table 4). For comparison, Egger et al. (2021) estimate a consumption-expenditure multiplier of 2.4 from cash

[47] We prefer medians to means, because the median may be less sensitive to networks with a high number of connections or many distant (low-*Closeness*) connections, where the linear specification for *Closeness* may be less appropriate. However, in our data the median and mean are in practice very similar.

transfers in Kenya, while in the United States, Nakamura and Steinsson (2014) estimate an "open economy relative multiplier" of 1.5, Suárez Serrato and Wingender (2016) estimate a local income multiplier of government spending of 1.7 to 2.0, and Chodorow-Reich (2019) suggest a spending multiplier of 1.8 based on a survey of multiple studies. Barrot and Sauvagnat (2016) find that one dollar of lost sales at the supplier level leads to \$2.40 of lost sales at the customer level. Relative to previous estimates of multipliers in developing countries (Egger et al. 2021), we exploit *within-village* variation in exposure to shocks based on distance to the shocked household in the village network. Thus, our estimates of indirect effects are net of any changes in prices (which would not differ by network distance). As such our multiplier estimate may be a lower bound, consistent with our estimate being at the lower end of the range of other recent estimates.

Because our calculations are based on economic responses of indirectly shocked households, *net* of changes in prices or other village aggregate adjustments, our multiplier estimates constitute novel partial equilibrium benchmarks for developing countries. The multiplier we estimate is, therefore, informative about the degree of frictions in insurance, labor, and goods markets.[48] While our multiplier estimates are admittedly back-of-the-envelope, they demonstrate that, because the indirect effects are economically meaningful and affect many households for each directly affected household, the total indirect effects are of a similar order of magnitude, and perhaps larger than, the direct effect itself.

B. *Network Structure and the Propagation of Shocks*

How does the structure of local networks affect propagation? Do shocks to agents who differ in their network position propagate to different extents? Understanding which economic microstructures are conducive to the amplification or attenuation of idiosyncratic shocks is relevant for targeting social protection (e.g., cash transfers), forecasting under what circumstances shocks will propagate to a greater or lesser extent, and gaining a deeper understanding of the ways in which networks function.

We shed light on these questions by estimating alternative specifications which, instead of exploiting variation in closeness to the shocked household *within* the village, use *cross-village* variation in shock exposure, combined with pre-post variation. We consider two dimensions of network exposure: the network's preshock density, and the degree centrality of the directly shocked household. Specifically, we estimate the following model:

$$
\begin{aligned}
(5)\quad y_{i,t,j} = {} & \beta Post_{t,j} \times NetworkExposure_j + \gamma Post_{t,j} \times Marketshare_j \\
& + \mathbf{X}_{i,t,j}\kappa + \theta_{\tau(j)} + \alpha_i + \omega_j + \delta_t + \delta_t \times NetworkExposure_j + \epsilon_{i,t,j},
\end{aligned}
$$

where, as above, the unit of observation is a household i in period t around the shock to household j. $NetworkExposure_j$, which does not vary across households i, measures

[48] The value of partial equilibrium multipliers is highlighted by Guren et al. (2021), who argue that partial equilibrium multipliers can be directly linked to specific mechanisms and be subject to a clearer interpretation as opposed to general equilibrium multipliers which capture multiple mechanisms.

either the shocked household's degree or the network's density during the preshock period, based on the transactions in the labor market and the market of inputs and final output. ω_j absorbs village-level variables that are invariant around the analysis window, including the main effect of $NetworkExposure_j$. We allow outcomes to vary over time differentially for villages with different structures by including time-fixed effects δ_t interacted with $NetworkExposure_j$. The vector $\mathbf{X}$ includes the interaction of the number of households in the village (number of nodes in the network) with $Post_{t,j}$ to account for potential contemporaneous shocks correlated with village size, as well as a measure of the household's market share in the local economy.[49]

The parameter of interest is β, which captures the post- versus preshock differences in changes in outcomes of households in villages with higher preshock network density (or shocked household degree centrality), relative to households in villages with lower preshock network density/shocked household degree centrality. It is useful to compare equation (4) with equation (5). Equation (4) leverages within-village variation, comparing households with a distant, or no, network connection to the shocked household versus those who are closer in the network. This specification traces out how the direct effects travel through the network. Equation (5) instead compares average outcomes of nondirectly shocked households in villages where network exposure differs; this specification sheds light on the indirect effects on the *average* nonshocked household, as a function of network characteristics.

The sign of β is theoretically ambiguous. When we consider variation in network density, on one hand, in denser networks the average household has more transaction partners, potentially making it easier to make up for the loss of transactions with the directly shocked household. On the other hand, being connected with more transaction partners increases overall indirect exposure to shocks, and may make shocks more "de facto aggregate." Similarly, a shock hitting a higher-centrality household may on one hand be better-insured or, due to homophily, the contacts of a higher-centrality household may also be more central/better insured; these mechanisms would suggest less propagation. On the other hand, a higher-centrality household has, by definition, more first-degree connections, which may cause the shock to propagate more.[50]

Results: Network Density, *Position, and Propagation*.—We first examine the effect of network density on shock propagation: panel A of Table 5 shows that shocks in denser networks propagate more: a 1 SD increase in the network's preshock network density results in 0.054 (5.4 percent) fewer input/output transactions in the postshock period, 0.027 (5.8 percent) fewer labor market transactions, and 0.08 (5.5 percent) fewer overall transactions (columns 1–3; all significant at the 1 percent level). The corresponding effect on income is a fall of 330 baht (3.1 percent) and the effect on consumption is a decline of 160 baht or 2.1 percent (columns 4 and 5), also statistically significant at 1 percent.

Turning to the effect of the network position of the shocked household, panel B in Table 5 indicates that when a more central household is shocked, greater

[49] Following Hulten (1978), we compute a firm's market share to be the firm's nonlabor revenues as a share of aggregate value added.

[50] Note that, because we are using only *network-level* variation in network density and in the centrality of the shocked household, we will not mechanically measure more propagation by virtue of more high-*Closeness* individuals, as we would if we used a specification like equation (3).

Table 5—Propagation and Network Characteristics

	Input/output (1)	Hired labor (2)	All transactions (3)	Total income (4)	Total spending (5)
Panel A. Aggregate shock amplification and network density					
Post × *Density* (z-score)	−0.054	−0.027	−0.080	−330.226	−160.752
	(0.014)	(0.014)	(0.022)	(94.026)	(33.558)
Observations	477,316	477,316	477,316	477,316	477,316
R^2	0.422	0.207	0.355	0.208	0.635
Preperiod mean	0.989	0.460	1.449	10,745	7,448
Number of events	410	410	410	410	410
Panel B. Aggregate shock amplification and shocked household degree					
Post × *Degree* (z-score)	−0.037	−0.046	−0.083	−273.657	−72.502
	(0.014)	(0.015)	(0.021)	(86.045)	(39.693)
Observations	477,316	477,316	477,316	477,316	477,316
R^2	0.422	0.207	0.355	0.208	0.635
Preperiod mean	0.989	0.460	1.449	10,745	7,448
Number of events	410	410	410	410	410

Notes: Panels A and B report results corresponding to equation (5) using network density as proxies of village-level exposure to shocks and degree centrality of the shocked household, respectively. All regressions include interactions of the postshock indicator with village size (number of households) and preshock Domar weights (market share) of the shocked household computed by dividing the shocked household gross revenues during the 12 months preceding the shock by the village-level aggregate value added during a similar time frame). Standard errors are clustered at the event level.

propagation results. A 1 standard deviation (SD) increase in the degree of the shocked household leads to an average of 0.037 (3.7 percent) fewer input/output transactions per household in the postshock period relative to the preshock period, 0.046 (10 percent) fewer labor market transactions, and 0.083 (5.7 percent) fewer overall per-household transactions (columns 1–3; all significant at 5 percent or better). Accordingly, a 1 SD increase in the degree of the shocked household leads to a differential fall in income of 273 baht (2.5 percent), significant at 1 percent (column 4). In column 5, the point estimate indicates a differential fall in consumption of 72 baht, or 0.9 percent, significant at 10 percent.

These results show that both the structure of the network and the location of the shocked household within it matter. Shocks to more centrally located households lead to greater propagation, as do shocks to denser networks, even after controlling for a household's market share in the local economy. These findings suggest a tradeoff in promoting network interlinkages: while increasing linkages among households may strengthen the insurance capacity of networks (Feigenberg, Field, and Pande 2013), such increased links may also decrease resilience to propagation. Likewise, encouraging links with central households may promote information diffusion (Beaman et al. 2021), but may increase indirect exposure to shocks via propagation as well. These results suggest that policy efforts targeted at attenuating the effects of shocks to central households may have high social returns, as they may prevent propagation. We explore this idea further in the next section.

C. *Access to Insurance and Attenuation of Propagation*

The above discussion illustrates features of the economic environment which mediate the propagation of shocks, namely network density and the position of the shocked household. It is also well known that insurance can help to buffer health shocks. Both formal insurance (Finkelstein et al. 2012) and informal risk-coping mechanisms (de Weerdt and Dercon 2006) may play a role by helping directly hit households buffer shocks. By allowing shocked households to cope *without* scaling back business activities, insurance can stop propagation before it starts.[51,52]

The Role of Informal Insurance.—We investigate the role of informal risk coping using heterogeneity in the strength of local insurance networks. We do so by allowing the direct and indirect effects of the shocks to vary based on whether the directly shocked household exhibits gift returns co-movements—a measure of engagement with risk sharing networks proposed by Samphantharak and Townsend (2018)—during the pre-period that are above or below the sample median.[53]

The results appear in online Appendix Table A13. Panel A reports direct effects.[54] Columns 1and 2 confirm that the severity of health shocks, as measured by symptoms or spending, does not differ by the baseline insurance network participation of the shocked household. Column 3 shows that the magnitude of incoming gifts/loans is roughly twice as high for high-insurance households (however the difference is not statistically significant). Reflecting this greater access to insurance, the falls in hired labor and business spending are roughly one-third as large for high-insurance households (columns 4 and 5). In order to examine heterogeneous effects with more statistical precision, column 6 reports on an inverse covariance weighted (ICW) index of labor and spending. The index falls by less than one-half as much, 0.05 versus 0.13 standard deviations (SD), for high- versus low-insurance households ($p = 0.0745$ for a one-sided test).[55]

Panel B presents the indirect effects, based on an specification similar to (5) but that allows the indirect effects of shocks hitting denser networks (a proxy for increased amplification of the shock) to vary by the shocked household's preshock engagement in insurance networks.[56] Both labor and input/output transactions, hence total transactions, fall to a lesser extent for indirectly exposed households when the directly shocked household has above-median access to informal insurance (columns 1–3). The falls in income and spending are also lower (columns 4

[51] Indeed, we document a role for informal insurance in our context, namely an increase in incoming gifts for directly shocked households in the first half year after the shock (see online Appendix Figure A5a). However, this increase does not fully make up for the sharp increase in health spending over the same time frame (see Figure 1, panel B). On average, the incoming gifts account for roughly two-thirds of the increase in health spending during the first postshock half year.

[52] The finding that directly shocked households see increases in net gifts and loans may raise the question of whether the indirect effect on consumption (see Table 2, column 6) could be a consequence of a decline in cash on hand/liquidity arising from helping the directly shocked household. However, online Appendix Table A12 shows that neither transfers nor loans given by the indirectly shocked household to other households increase following the shock.

[53] Because Sections IVC and IVD together estimate three different sets of heterogeneity analyses for direct effects and a corresponding effect for indirect effects, one may worry that some of the effects will be spuriously significant. Therefore, for each set of heterogeneity analyses, we construct an inverse covariance weighted index of key outcomes and compute Anderson (2008) q-values across the three indices for direct effects and the three indices for indirect effects. These are reported in online Appendix Tables A13, A14 and A17.

[54] Specifically, we estimate direct effects by estimating β_1 and β_2 from equation (B6), which is a modified version of (2). Details appear in online Appendix B.B6.

[55] We discuss one-sided p-values in this section in keeping with the one-sided alternative that better access to insurance *mitigates* direct and indirect effects of shocks. Online Appendix Tables A13–A17 also report two-sided p-values.

[56] We estimate indirect effects by estimating β_1 and β_2 from equation (B7), which is a modified version of equation (5). Details appear in online Appendix B.B6.

and 5). An ICW index of total transactions, income and spending falls by 0.023 versus 0.038 SD when the shocked household is well insured versus not ($p = 0.086$). In sum, when shocks hit households with better access to informal insurance, both the direct and indirect effects are mitigated. This result emphasizes the distinct roles of economic networks: production networks (e.g., supply chains and employer-employee networks) appear to amplify idiosyncratic shocks but insurance networks contribute to attenuate their effects.

The Role of Formal Insurance.—To explore the role of formal insurance in mediating direct and indirect effects of shocks, we explore heterogeneity across the four provinces in our sample. As documented by Gruber, Hendren, and Townsend (2014), Thailand's "30 Baht" program, which increased funding available to hospitals to care for the poor and reduced copays to 30 baht ($0.75), was implemented more intensively and was more impactful in poorer provinces.[57] Thus, we examine whether the direct and indirect effects of shocks differ between the poorer provinces in the sample (Buriram and Sisaket), which were more intensively targeted by the 30 Baht program, versus the richer provinces (Chachoengsao and Lopburi). Of course, there are many other important differences across provinces. While *level* differences across provinces or villages will be absorbed by the household fixed effects, this analysis should be regarded as suggestive.

Online Appendix Table A14 presents the results. Direct effects are presented in panel A. Despite health shocks in poorer provinces being slightly more severe as measured by symptoms (column 1), the rise in health spending is less than half as large (column 2, $p < 0.01$). Given that the shocks require less out of pocket expenditure, incoming gifts/loans are smaller in poorer provinces (column 3). Columns 4 and 5 show that in the poorer provinces, where formal health insurance was more robust, the falls in both hired labor and business transactions are roughly one-third the size as in other provinces. Accordingly, the index of labor and spending falls by 0.0362 SD in poorer provinces versus 0.128 SD in richer provinces ($p = 0.062$).

Panel B presents the indirect effects. Transactions (labor, goods, and total) fall to a lesser extent for indirectly shocked households in poorer provinces (columns 1–3). Accordingly, income and consumption also fall to a lesser extent. An index of total transactions, income and spending falls by 0.086 SD in richer provinces versus 0.031 SD in poorer provinces ($p < 0.01$). Thus, in areas where formal insurance was implemented more robustly, direct effects of health shocks are buffered and therefore, propagate to a lesser extent.

D. *The Role of Labor Market Incompleteness*

Frictions in the market for labor may drive the direct—and in turn, indirect—effects of shocks. The lost labor of an ill or injured household member (and other members taking care of them) may not easily be replaced by hired labor. We find declines in hired labor coupled with similar declines in labor provided by household members among directly shocked households (columns 6–7 of Table 1), which are not offset

[57] A key aim of the policy was to reduce geographical disparities in the provision of public healthcare.

with incoming free external labor (column 5, panel A of online Appendix Table A15). Moreover, online Appendix Table A16 shows that when a shock hits working-age members (those more likely to operate businesses) there is a large decline in hired labor and a corresponding decline in business spending. This suggests that there may be complementarities between labor provided by household members and hired labor and hence, when household labor declines due to a shock, demand for hired labor will fall as well.

To further investigate to what extent demand-side frictions in local labor markets mediate propagation, we split the sample by the degree of complementarity between household and hired labor, $c_i^{h,l}$, measured as the preshock co-movement between the idiosyncratic component of labor provided by household members (h) and labor hired externally (l). If labor market frictions play a role, we expect to see larger effects for those households with above-median $c_i^{h,l}$ (see online Appendix Section B.B6 for details).

Panel A in online Appendix Table A17 first shows that the severity of the shock, measured via symptoms and by spending on health, is not different for high- versus low-$c_i^{h,l}$ households (columns 1and 2). The degree of informal insurance is also similar (column 3). However, the fall in hired labor is over 6 times larger for high-$c^{h,l}$ households (column 4) $(p = 0.0734)$. In contrast, the direct effects on business spending are similar (column 5), indicating that the difference is specific to labor, not goods, demand.

However, turning to the indirect effects (panel B), there is little evidence of heterogeneity by the degree of complementarity $(c_i^{h,l})$ of the shocked household: the falls in transactions, and consumption are similar when high- and low-$c_i^{h,l}$ households are shocked. Although the indirect effects on income appear to differ (though not significantly), the impacts on the ICW index of total transactions, income and spending are remarkably similar. The results, in sum, suggest that while labor market frictions play a role in shock propagation, as shown by the greater effects of labor market versus supply chain exposure in Table 2 panel B, the degree of internal-external labor complementarity of the shocked household is not a key mediator of these indirect effects. This may indicate that the predominant channel for propagation is the expenditure-side impacts of health shocks combined with supply-side frictions in local labor markets (e.g., costs of migrating outside of the village), rather than impacts to household labor endowments and attendant frictions on the demand for hired labor.

V. Concluding Remarks

In developed economies, firms are often owned by diversified investors and embedded in global supply chains. In contrast, small firms in LMICs are often operated by un-hedged households and are tightly linked to other businesses and households through local economic networks defined by labor as well as product market transactions. These key differences may imply that the way in which firms cope with shocks to their owners, and the mechanisms through which these shocks propagate through the local economy may also be different to the behavior documented in the literature studying larger firms.

Our analysis reveals three novel facts. The first is that the adjustments that households make to cope with severe idiosyncratic shocks not only affect their business outcomes, but also the incomes and spending levels of other households. In our setting, health shocks put substantial pressure on household budgets and labor endowments. As a result, shocked households adjust production: drawing down working capital, cutting input spending, and reducing labor hiring. These adjustments propagate the shock to other households through interlinkages in local supply chain and labor market networks. The aggregate indirect effects imply a consumption multiplier of approximately 1.23.

The second fact we document is that, while these shocks propagate through both local supply chain and local labor markets, propagation through labor linkages is more consequential in terms of indirect effects on consumption. This distinction points at important labor market frictions: once a link between and employer and an employee is broken, it is quite hard to replace it, either locally or by selling/buying labor outside the network.

Thirdly, we document that economic microstructures also shape propagation: shocks in denser networks propagate more, as do shocks to actors who are central in networks. Additionally, access to insurance (either formal or informal) mediates the spread of shocks, by decoupling the strategies that households use to cope with shocks and their productive decisions. As such, insurance appears to mitigate propagation by "stopping it before it starts."

Our findings suggest several interventions that may be beneficial. First, investing in preventative heath to reduce the severity of shocks benefits not only the household whose health improves, but also others linked in the local network. In addition, improved safety nets may help cut the link between production and the strategies that business owners use to cope with shocks, preventing granular shocks from propagating. Safety nets such as workfare (e.g., India's National Rural Employment Guarantee Scheme), health insurance (e.g., Thailand's 30 Baht Program), cash transfers (e.g, Mexico's Oportunidades), and others benefit not only the directly targeted household but also its network connections; either by reducing propagation or by reducing its negative second-order effects. Second, where such programs are not universal, they may benefit from targeting those with key roles in the underlying networks, or areas (networks) where the economic activities of their members are more interconnected (i.e., higher network density). Electronic payment platforms that identify key players in the network structure could allow insurers to better target recipients who are key nodes in local networks.

Finally, policy interventions should also strive to improve the functioning of local labor markets and to make production networks less rigid and more diversified. Interventions to improve contract enforcement (Fazio, Silva, and Skrastins 2020) or to broaden the extent of product and factor markets beyond the local village market (Park, Yuan, and Zhang 2021) may reduce the rigidity and sparsity of local supply chain and labor market networks and hence mitigate the propagation and persistent adverse impacts of idiosyncratic shocks.

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